

Human Vision – Topic 3

Computational Approach to Perception of Color

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May 2, 2026

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1 Lateral Inhibition and Contrast

1.1 The Phenomenon: Simultaneous Contrast

Simultaneous contrast is a classic visual phenomenon where the perceived brightness of an object is heavily influenced by the luminance of its immediate surroundings.

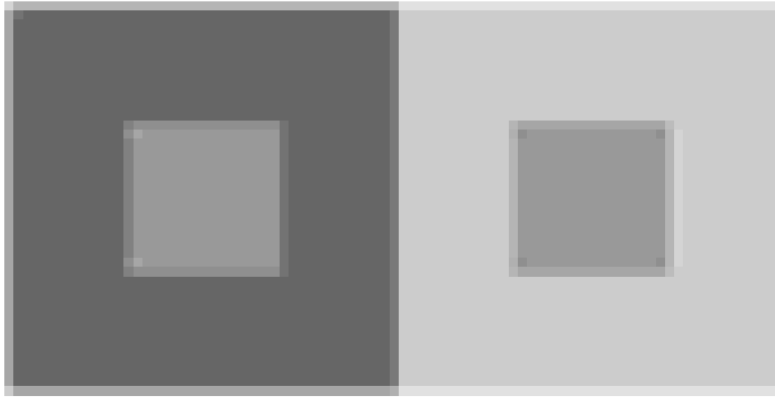


Figure 1: Two identical gray squares placed against dark and light backgrounds.

In Figure 1, although both central squares are objectively the exact same shade of gray, the square on the darker background appears lighter, while the square on the lighter background appears darker.

1.2 The Mechanism: Center-Surround Receptive Fields

To understand why simultaneous contrast happens, we have to look at how individual neurons early in our visual system (like the retinal ganglion cells in our eyes) process light.



Figure 2: A center-surround receptive field showing an excitatory center (+) and an inhibitory surround (-).

As shown in Figure 2, these neurons act like tiny contrast detectors. Their firing pattern depends on the arrangement of light across two opposing regions:

- **The Center (+):** When light hits this middle area, it excites the neuron, causing it to fire more.

- **The Surround (-):** When light hits this outer ring, it inhibits (slows down) the neuron's firing.

Ultimately, the neuron fires based on the **difference** (subtraction) between the light hitting the center and the light hitting the surround.

Clarification: What is the "Center"?

The "center" does **not** mean the center of the object you are looking at.

- Think of human vision like a digital camera sensor made of millions of tiny pixels. Every individual neuron has its own microscopic "receptive field" (its own pixel).
- The **center** is just the middle of that one specific neuron's field of view.
- Our eyes are covered in a mosaic of millions of these tiny center-surround detectors.

1.3 Explaining the Illusion

We can easily explain the simultaneous contrast illusion by applying this "subtraction" math to the backgrounds:

- **Dark Background:** The center (+) sees the gray square and wants to fire. The surround (-) sees the black background, which has almost no light to subtract. The signal stays strong, so the brain perceives the square as **lighter**.
- **Light Background:** The center (+) sees the exact same gray square. However, the surround (-) is flooded with light from the white background. It subtracts a large amount from the signal, making the brain perceive that same gray square as much **darker**.

1.4 Successes of the Theory: The Vasarely Illusion

While the lateral inhibition model dominates much of the computational thinking around brightness perception, it is important to note that it cannot perfectly explain *every* visual phenomenon. However, there are striking examples where the theory aligns flawlessly with our perception.

The first being the classic illusion by the artist Victor Vasarely.



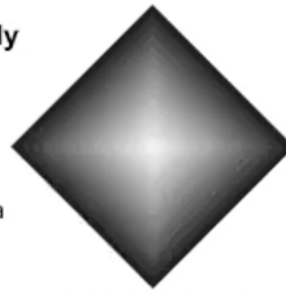
An Illusion by Victor Vasarely

We see a bright "cross" within this diamond. However, the arms of the cross are no brighter than the diamonds on which they rest. This illusion may result from the lateral inhibition occurring in the retina and other stages of early visual processing.

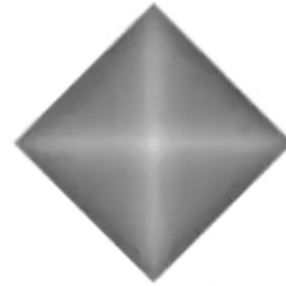


the receptive field of an idealized center-surround cell

An array of cells with center-surround organization can be considered to filter the top image. The output of such cells is shown in the bottom image, where the bright lines of the cross are really there.



An illusion by Vasarely



center-surround filtered

Figure 3: The Victor Vasarely Illusion. Nested, uniformly shaded squares create the false perception of a glowing cross.

When we stack multiple uniformly-shaded squares of decreasing size on top of one another, we perceive a bright, glowing "X" or cross radiating from the center of the diamond out to the corners. In reality, the arms of this cross are no brighter than the flat edges of the diamonds on which they rest.

This illusion is a direct result of lateral inhibition occurring in the retina and other stages of early visual processing.

The Geometry of the Glowing Cross

We can explain this glowing effect using the spatial geometry of our center-surround receptive fields:

- **On a straight edge:** When a neuron's receptive field sits on the flat edge of a square, the inhibitory surround (-) overlaps evenly with the darker outer square, subtracting a standard amount of light.
- **At a corner:** When the neuron sits precisely in the sharp corner of a square, the geometry changes. The inhibitory surround (-) now hangs over *more* of the darker outside area.

Because the surround is looking at more darkness at the corners, it **subtracts less** from the center's signal. Less subtraction means a higher neural output, which tricks the brain into perceiving the corners as brighter than the flat edges.

1.5 The Craik-O'Brien-Cornsweet Effect

Another powerful demonstration of lateral inhibition and our visual system's reliance on edges is the **Craik-O'Brien-Cornsweet Effect**.

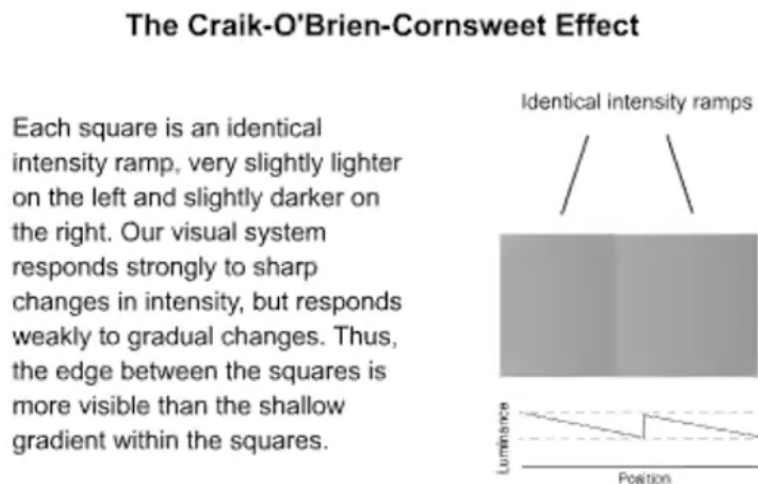


Figure 4: The Craik-O'Brien-Cornsweet Effect. The sharp central edge dictates the perceived brightness of the two identical gradients.

In Figure 4, the left rectangle appears uniformly darker than the right rectangle. In reality, both shapes are identical **intensity ramps** (gradients). They are slightly lighter on the left and slowly become darker on the right.

The "Filling-In" Process

We can explain this using the behavior of center-surround receptive fields:

- **High response to sharp edges:** At the boundary where the two shapes meet, there is a sudden, sharp jump in luminance. Our center-surround neurons are highly sensitive to this abrupt contrast, firing strongly to signal a distinct border.
- **Low response to gradual changes:** Across the rest of the shape, the luminance changes incredibly slowly. Because the difference in light between a neuron's center and its surround is negligible during a slow gradient, the neurons barely fire, signaling uniform brightness.

Because the visual system receives almost no contrast data from the gradual slopes, the brain relies entirely on the strong signal from the sharp edge. It effectively "fills in" the rest of the surface based on that edge, tricking us into seeing two flat, uniformly shaded squares.

2 Limitations of Lateral Inhibition

While lateral inhibition successfully explains low-level edge effects, there are many phenomena it fails to explain. Other examples, such as "snake" or "haze" lightness illusions, show that simple inhibition is not a sufficient explanation. In these cases, overall spatial configuration and higher-level interpretations of 3D form override the local "subtraction" math of early neurons.

2.1 The Koffka Ring

The Koffka Ring is a classic example demonstrating that grouping can overpower local contrast.

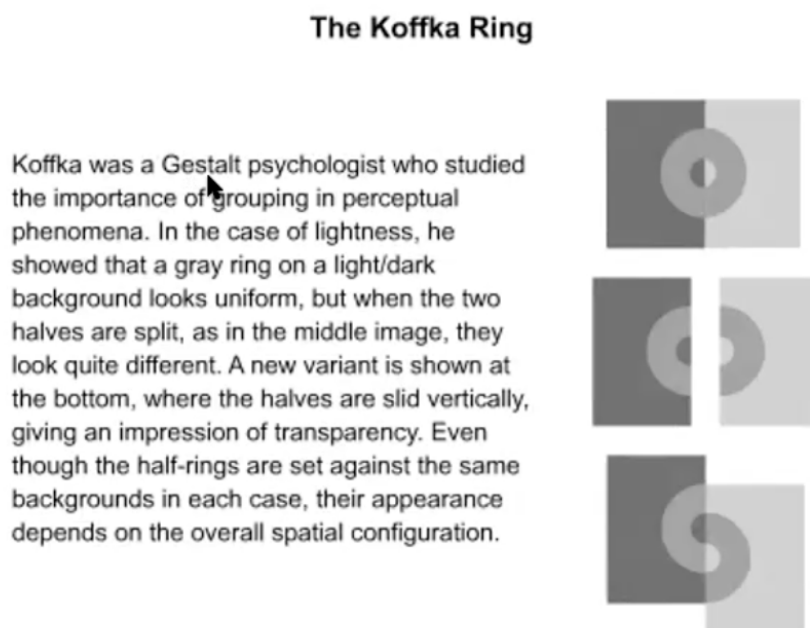


Figure 5: The Koffka Ring. The appearance of the gray shape depends on its overall spatial configuration.

According to the strict theory of lateral inhibition, the intact gray ring (top image) should appear as two different shades because its two halves rest on dramatically different backgrounds (simultaneous contrast). However, to the human eye, the intact ring appears as a uniform shade of gray.

The Failure of Local Contrast

It is only when the ring is physically split (middle image) that the two halves appear to be different shades. This proves that lateral inhibition is not the only mechanism at play; when the shape is grouped as a single, continuous object, the brain suppresses the local contrast illusion.

2.2 Knill and Kersten's Illusion

Similarly, Knill and Kersten demonstrated that the powerful Craik-O'Brien-Cornsweet Effect can be severely weakened by our brain's higher-level perception of 3D shapes.

Knill and Kersten's Illusion

This illusion shows that the Craik-O'Brien-Cornsweet effect can be influenced by higher level aspects of form perception. Both figures are the same pair of COCE squares. In the top case, the context suggests that both squares are in the same plane and one is painted a darker color. In the bottom case, the two squares are interpreted as shaded cylinders. In this image, there is thus no need to "explain" the step edge in terms of paint. As a result, the illusion is weaker.



Figure 6: Knill and Kersten's Illusion. The COCE is strong in the top flat shapes but weak in the bottom cylinders.

Both the top and bottom objects in Figure 6 are constructed using the exact same pair of COCE gradient squares.

- **Top Case (Flat Cubes):** The visual context suggests these are flat surfaces on the same geometric plane. The brain interprets the sharp step edge as a change in *paint color*, maximizing the COCE illusion.
- **Bottom Case (Cylinders):** The visual context (the curved tops and bottoms) forces the brain to interpret the shapes as 3D cylinders. Because cylinders naturally have shading gradients due to lighting, the brain explains away the luminance ramp as *shadows* rather than a change in paint.

As a result, there is no need for the visual system to "fill in" the color from the edge, and the illusion is heavily diminished.

3 The Gestalt Approach

As demonstrated by the Koffka Ring and Knill & Kersten's cylinders, early neural processing is only part of the story. The visual system does not just passively calculate local light differences; it actively interprets the entire image.

The **Gestalt approach** argues that human perception is fundamentally based on the **organization of the scene into objects, surfaces, and lighting conditions**. Rather than analyzing an image neuron-by-neuron, the brain uses global context, grouping principles, and 3D interpretation to figure out what we are actually looking at. In this view, "the whole is greater than the sum of its parts."

4 Computational Approach

To truly understand perception, we must ask: what is the fundamental problem the visual system is trying to solve? Why is it difficult, and what constraints make it possible to solve? To answer this, we take a detour into the physics of light.

4.1 Lambertian Reflectance

The Lambertian Law is a model describing how light reflects off a surface. It states that the amount of light reaching our eyes (I) is a function of three variables:

$$I = \rho L \cos(\theta)$$

Where:

- I is the image intensity (the light reaching the eye).
- ρ (rho) is the **albedo** (the fraction of light reflected, or the "paint color").
- L is the intensity of the light source.
- θ (theta) is the angle between the surface normal and the light source.

This physical model holds true for **matte objects** (e.g., a plaster wall), as opposed to shiny/specular objects. A key feature of Lambertian surfaces is that they appear equally bright from any viewing angle; the amount of light reaching the camera only depends on those three parameters, not the camera's position in the scene.

4.2 The Ill-Posed Problem of Inverse Optics

Why is visual perception mathematically difficult?

When we operate in the physical world, we want to know an object's true properties—for example, whether a person is wearing a black shirt or a white shirt. What we are actually looking for is ρ (the albedo). A black shirt reflects a small amount of light ($\rho \approx 0.1$), while a white shirt reflects much more ($\rho \approx 0.7$), and snow reflects even more ($\rho \approx 0.9$).

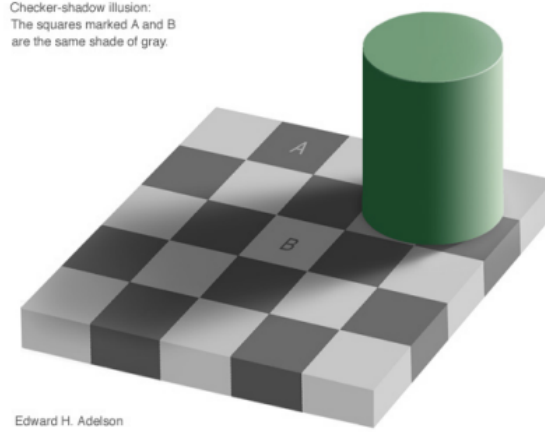
However, the eye does not receive ρ . It only receives I (the product of the three numbers). We are given I , but we need to estimate all three underlying components (ρ , L , and θ).

An Impossible Math Problem

Mathematically, you cannot solve a single equation with three unknown variables. This is known as an **ill-posed problem**. Yet, our visual system is exceptionally good at solving it. When lighting or geometry changes in a scene, we intuitively know exactly what changed—the paint, the shape, or the shadow.

4.3 Solving the Problem: Adelson's Checker-Shadow Illusion

Let's revisit how the brain solves this using Edward Adelson's famous Checker-Shadow illusion.



$$I = \rho L \cos(\theta)$$

Figure 7: Adelson's Checker-Shadow Illusion. Squares A and B send the exact same amount of light to the eye.

In Figure 7, Cell A and Cell B reflect the exact same amount of light to our eyes ($I_A = I_B$). Yet, our visual system perceives them as entirely different colors (A looks dark, B looks light).

Why does the brain do this? Because it is trying to isolate the albedo (ρ).

According to the Lambertian equation:

$$I_A = \rho_A L_A$$

$$I_B = \rho_B L_B$$

The brain uses global context to determine that Cell B is inside the shadow cast by the cylinder. Therefore, it estimates that the illumination hitting B is much lower than the illumination hitting A ($L_A > L_B$).

The Logic of the Illusion

If $I_A = I_B$, but the brain knows $L_A > L_B$, then mathematically, it must be true that:

$$\rho_A < \rho_B$$

Our visual system concludes that Cell A is painted with a darker pigment (ρ_A) than Cell B (ρ_B). The illusion is not a failure of our eyes; it is a successful demonstration of our visual system solving the inverse optics problem by factoring in illumination!